GEOSOFTWARE.ID
EDUCATION - DEVELOPMENT - INNOVATION

CERTIFICATE OF COMPLETION

Certificate Number: A.C.1532/WebGIS-Py/X/2025

This is to certify that :

Muhammad Sayyid Murtaja

Has Successfully Completed the Training Titled :

WebGIS Fundamentals with Python: From Scratch to Online Publication via the Web

Conducted on October 10, 11, 12, 2025

Total duration : 540 Minutes (12 JPL)

Geosoftware.ID

Ulvienin Harlianti, ST., MT.
Director of Geosoftware.ID

Ardianto
Instructor at Geosoftware.ID

Sesi	Topik	Materi	JPL
Session 1 Friday, October 10, 2025	Technical Foundations & Introduction to Folium	• Python installation • Creating a virtual environment • Introduction to Visual Studio Code & extensions • Basic use of Python for WebGIS • Introduction to Folium & creating a basic interactive map • Adding geometric objects (markers, polylines, polygons, popups, tooltips) • Integrating external data (CSV, GeoJSON, SHP) • Enhancing map interactivity with Folium plugins	4
Session 2 Saturday, October 11, 2025	Web Application Development with Flask & Frontend	• Basic HTML structure and elements • Styling pages with CSS (selectors, layout, box model) • Applying Bootstrap (grid system, utilities, UI components) • Creating responsive web pages using Bootstrap components	4
Session 3 Sunday, October 12, 2025	WebGIS Integration & Deployment	• Basic concepts of client-server & HTTP • Introduction to Flask (routing, templates, data passing) • Building a complete WebGIS application with Flask & Folium • Adding data filtering features & dynamic views using templates • Integrating external data in a WebGIS application • Application deployment concepts • Using a GitHub repository • Deployment with Vercel	4
Total Seluruh JPL			**12**